Linux P2P Ransom Botnet Investigation - Student Briefing

1. Malware Overview

- Family / type: Linux P2P Ransom botnet (Rex)
- Behaviour class: Ransomware with botnet functionality
- What it does:
- Spreads through peer-to-peer networks
- Encrypts files on infected Linux systems
- Demands ransom for decryption
- Has command-and-control (C2) capabilities
- Participates in botnet activities
- Associated MITRE ATT&CK techniques:
- T1486 Data Encrypted for Impact
- T1071 Application Layer Protocol
- T1059 Command-Line Interface
- T1053 Scheduled Task/Job
- T1078 Valid Accounts
- T1021 Remote Services
- T1041 Exfiltration Over C2 Channel
- T1055 Process Injection
- T1057 Process Hollowing
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- T1087 Account Discovery
- T1089 Disk Wipe
- T1112 Modify Registry
- T1113 Screen Capture
- T1119 Automated Exfiltration
- T1120 Peripheral Device Discovery
- T1123 Audio Capture
- T1124 System Information Discovery
- T1125 Video Capture
- T1127 Triggered Email Collection
- T1129 Internal Spearphishing
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- T1133 External Remote Services
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- T1500 System Location Discovery
- Associated CVEs: Not specified in this scenario

2. Exercise Objective

- The learning goal of this cyber-range exercise is to investigate a Linux P2P Ransom botnet malware attack on a victim machine, identify indicators of compromise, and take appropriate actions to mitigate the threat.
- Difficulty level: Intermediate
- Cybersecurity approach: Defence-oriented
- NICE work role: Incident Response
- By the end of this exercise, a successful trainee will have demonstrated the ability to identify and analyze ransomware behavior, extract C2 server information, and understand the steps involved in mitigating a ransomware attack.

3. Sandbox Topology & Malware Emulation

Sandbox Topology

```
192.168.90.0/24
  |
  |-- router (192.168.90.1)
  |-- c2-server (192.168.90.5)
  |-- victim (192.168.90.10)
  |-- exfil-server (192.168.90.20)
```

- Hosts:
- c2-server: Command and control server (192.168.90.5)
- victim: Infected machine (192.168.90.10)
- exfil-server: Exfiltration server (192.168.90.20)
- router: Network router (192.168.90.1)

Malware Emulation

- Ransomware Key Generation:
- Host: victim
- Artifact: /root/.ransom_key_backup (stores the AES-256 encryption key)
- Log: /var/log/ransom-transmission.log (simulates C2 beacon)
- Key Server:
- Host: c2-server

- Artifact: /root/.ransom-keys/victim_key.txt (stores the victim's decryption key)
- Service: ransom-keysrv.service (listens on port 4443)
- Log: /var/log/ransom-c2/keysrv.log (logs key server activity)
- Exfiltration:
- Host: victim
- Artifact: /var/log/ransom-exfil.log (logs exfiltration completion)
- Command: rsync (used to exfiltrate data to exfil-server:/var/exfil/victim/)
- Encryption:
- Host: victim
- Artifact: /tmp/.crypt_agent (ransomware script left behind by the attacker)
- Log: /var/log/ransom-encrypt.log (logs encryption events)
- Ransom Note: /home/cyrano/RANSOM_NOTE.txt (contains instructions for decryption)

4. Student Tasks

ACCESS Step

- Credentials:
- Username: cyrano
- Password: cyrano
- Start Passkey: start

TRAINING_LEVEL 1: Identify C2 Server Port

- Task: Read the ransom note left by the attackers and identify the C2 server port number.
- Expected Artifact Type: Port number

TRAINING_LEVEL 2: Identify Ransomware Script

- Task: Find the ransomware script left behind by the attacker that describes the steps performed by the ransomware malware.
- Expected Artifact Type: File path

TRAINING_LEVEL 3: Identify Exfiltration Destination

- Task: Analyze the ransomware script to find the remote destination used to send files to the exfiltration server.
- Expected Artifact Type: Exfiltration destination path

5. Solution Walkthrough & Flags

TRAINING_LEVEL 1: Identify C2 Server Port

- Methodology:
  1. SSH into the victim machine using the provided credentials:
     ```
     ssh cyrano@192.168.90.10
     ```
  2. Navigate to the home directory of the user cyrano:
     ```
     cd /home/cyrano
     ```
  3. Read the contents of the file RANSOM_NOTE.txt:
     ```
     cat RANSOM_NOTE.txt
     ```
  4. Identify the C2 server port number mentioned in the ransom note.
- Expected Flag: 4443

TRAINING_LEVEL 2: Identify Ransomware Script

- Methodology:

1. SSH into the victim machine using the provided credentials:
   ```
   ssh cyrano@192.168.90.10
   ```
2. Navigate to the /tmp directory:
   ```
   cd /tmp/
   ```
3. Look for hidden files using the ls -la command:
   ```
   ls -la
   ```
4. Identify the ransomware script.

- Expected Flag: /tmp/.crypt_agent

TRAINING_LEVEL 3: Identify Exfiltration Destination

- Methodology:
1. SSH into the victim machine using the provided credentials:
   ```
   ssh cyrano@192.168.90.10
   ```
2. Navigate to the /tmp directory:
   ```
   cd /tmp/
   ```
3. Read the contents of the ransomware script identified in the previous step:
   ```
   cat /tmp/.crypt_agent
   ```
4. Identify the remote destination used for exfiltration.

- Expected Flag: exfil-server:/var/exfil/victim/

6. Knowledge Gained

- Skills Practiced:
- Investigating ransomware attacks
- Identifying indicators of compromise
- Analyzing ransomware behavior
- Extracting C2 server information
- Understanding exfiltration mechanisms
- Tools Used:
- SSH
- cat command
- ls -la command
- Concepts Learned:
- Ransomware infection vectors
- Command-and-control (C2) capabilities
- Exfiltration techniques
- Persistence mechanisms
- NICE Role Alignment: Incident Response
- MITRE Techniques Alignment:
- T1486 Data Encrypted for Impact
- T1071 Application Layer Protocol
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